

CSC3002F - Operating Systems II

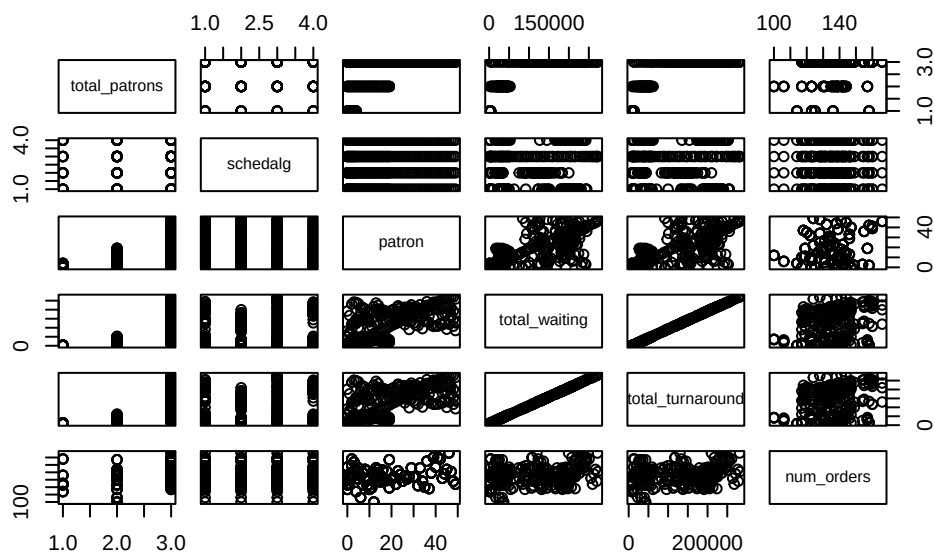
Scheduling Assignment

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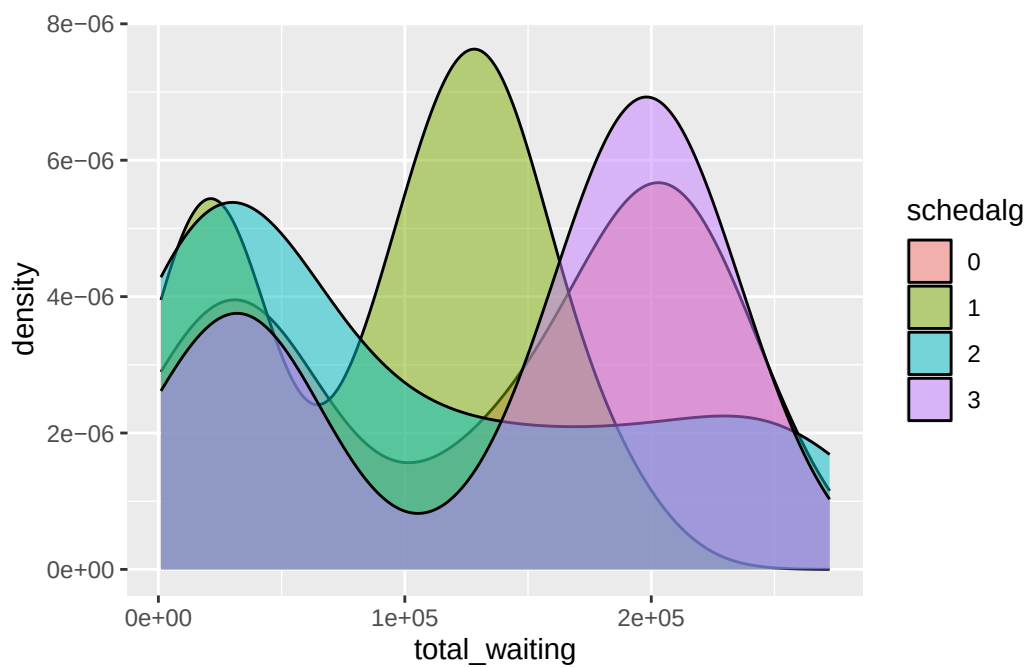
17th May 2026

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```
patron_summary <- df %>%
  group_by(total_patrons, schedalg, patron) %>%
  summarise(
    total_waiting = sum(waiting_time),
    total_turnaround = sum(turnaround_time),
    num_orders = n(),
    .groups = 'drop'
  )
plot(patron_summary %>% group_by(total_patrons, schedalg))
```

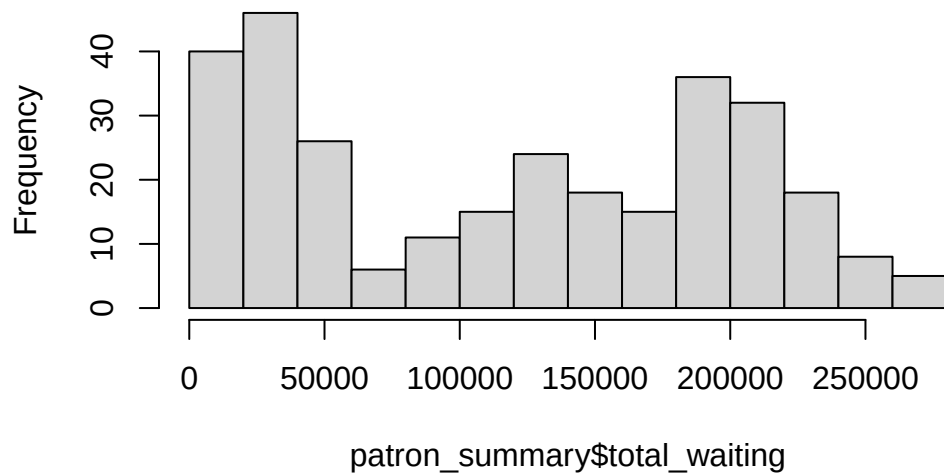


```
ggplot(patron_summary,aes(x=total_waiting,fill=schedalg)) + geom_density(alpha = 0.5)
```

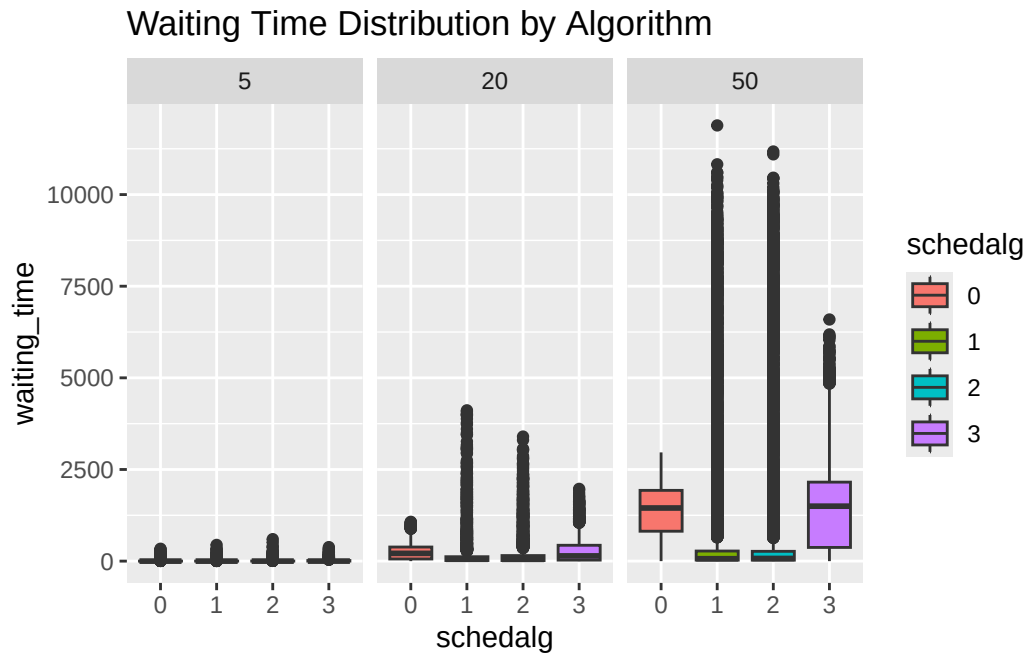


```
hist(patron_summary$total_waiting)
```

Histogram of patron_summary\$total_waiting



```
ggplot(df, aes(x=schedalg, y=waiting_time, fill=schedalg)) +  
  geom_boxplot() +  
  facet_wrap(~total_patrons) +  
  labs(title="Waiting Time Distribution by Algorithm")
```



```
# 1. Fit the model
# We use factor() for sched_alg because it's a category, not a continuous number
fit <- lm(waiting_time ~ factor(schedalg) + total_patrons + drink, data = df)

# 2. View the significance (p-values) and magnitudes (coefficients)
summary(fit)
```

Call:

```
lm(formula = waiting_time ~ factor(schedalg) + total_patrons +
    drink, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-2325.0	-833.6	-155.8	297.6	10117.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1090.57	37.88	28.788	< 2e-16 ***
factor(schedalg)1	-295.37	18.80	-15.711	< 2e-16 ***
factor(schedalg)2	-230.18	18.80	-12.244	< 2e-16 ***
factor(schedalg)3	61.31	18.80	3.261	0.00111 **

total_patrons20	206.03	29.20	7.057	1.73e-12	***
total_patrons50	1174.16	27.38	42.883	< 2e-16	***
drinkBeer	-1104.22	36.60	-30.172	< 2e-16	***
drinkBloody Mary	-1071.83	36.71	-29.197	< 2e-16	***
drinkCider	-1054.76	35.58	-29.648	< 2e-16	***
drinkCosmopolitan	-1126.20	37.48	-30.045	< 2e-16	***
drinkDraft Beer	-1132.50	36.22	-31.269	< 2e-16	***
drinkGin and Tonic	-1111.39	36.48	-30.462	< 2e-16	***
drinkMargarita	-866.04	36.60	-23.664	< 2e-16	***
drinkMartini	-1115.89	36.04	-30.967	< 2e-16	***
drinkMojito	-824.73	35.86	-22.997	< 2e-16	***
drinkPina Colada	-484.36	36.00	-13.455	< 2e-16	***
drinkRed Wine	-1092.50	35.73	-30.577	< 2e-16	***
drinkTequila shot	-1132.41	37.16	-30.473	< 2e-16	***
drinkVodka shot	-1094.05	36.16	-30.253	< 2e-16	***
drinkWhite Wine	-1101.26	36.71	-29.998	< 2e-16	***

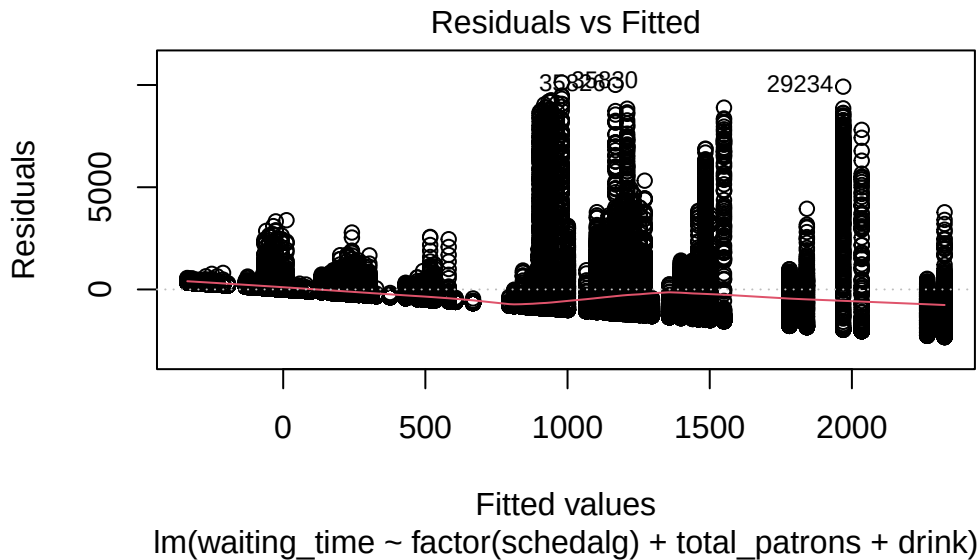
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1339 on 40592 degrees of freedom

Multiple R-squared: 0.1611, Adjusted R-squared: 0.1607

F-statistic: 410.2 on 19 and 40592 DF, p-value: < 2.2e-16

```
# 3. Check for Fairness/Starvation by plotting residuals
plot(fit, which = 1)
```



```
# Compare means
model_speed <- lm(waiting_time ~ factor(schedalg) + total_patrons, data = df)
summary(model_speed)
```

Call:

```
lm(formula = waiting_time ~ factor(schedalg) + total_patrons,
    data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1375.9	-955.4	-133.9	242.5	10867.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	133.95	29.31	4.570	4.88e-06	***
factor(schedalg)1	-295.37	19.29	-15.316	< 2e-16	***
factor(schedalg)2	-230.18	19.29	-11.936	< 2e-16	***
factor(schedalg)3	61.31	19.29	3.179	0.00148	**
total_patrons20	211.40	29.94	7.060	1.69e-12	***
total_patrons50	1180.63	28.08	42.042	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1374 on 40606 degrees of freedom

Multiple R-squared: 0.1169, Adjusted R-squared: 0.1168

F-statistic: 1075 on 5 and 40606 DF, p-value: < 2.2e-16

```
plot(model_speed)
```

